Dusa Gouresh

co24btech11007@iith.ac.in | gouresh.dusa@gmail.com | +91 9833646382 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.99
Sri Chaitanya Jr. College	Class XII (TSBIE)	2024	98.40%
Dr. Antonio da Silva High School	Class X (MSBSHSE)	2022	93.80%

PROJECTS

Customer Churn Predictor (Machine Learning)

Personal Project | Python, Scikit-Learn, Flask, HTML/CSS

- **Predictive Modeling:** Built and trained a Random Forest classification model in Python to accurately predict customer cancellation risks based on tenure, billing, and support history.
- **Full-Stack Integration:** Developed a seamless backend using Flask to connect the machine learning model directly to a clean, interactive web interface built with HTML and CSS.
- **Business Application:** Designed the application to instantly translate raw user data into easy-to-read probability scores, providing actionable insights for management to improve customer retention.

Enhanced xv6 Operating System

Course Project | C, QEMU, Linux, RISC-V

- **Kernel & Scheduling:** Extended xv6 RISC-V kernel (C) with a 4-level SC-MLFQ CPU scheduler.
- **Virtual Memory:** Built a memory subsystem with lazy allocation and Clock page replacement.
- **Storage Systems:** Implemented disk-backed swapping, FCFS/SSTF scheduling, and RAID storage.
- **System Calls:** Added custom system calls for deep process accounting and memory tracking.

Computational PDE Solver (Numerical Methods)

Course Project | C, Python, Numerical Analysis, Partial Differential Equations (PDEs)

- **Numerical Implementation:** Developed high-performance finite difference solvers in **C** for 1D/2D advection, diffusion, and non-linear Burgers' equations.
- **Algorithmic Optimization:** Engineered an **Alternating Direction Implicit (ADI)** scheme using the **Thomas Algorithm (TDMA)** for efficient tridiagonal matrix inversion in 2D systems.
- **Analysis & Visualization:** Performed rigorous stability (von Neumann) and convergence analysis across varying Péclet regimes; utilized **Python (NumPy/Matplotlib)** to visualize complex wave propagation and shock formation.

SKILLS

Programming Languages : C, C++, Python, SQL

Tools: Git, GitHub, VS Code, Linux Command Line, Overleaf

Libraries: NumPy, Matplotlib

POSITIONS OF RESPONSIBILITY

Competitions Head | Elan & Vision, IIT Hyderabad *(2026-Present)*

- Leading a team of **13 coordinators** to plan and execute **30+ technical, cultural, and literary competitions** for IIT Hyderabad's annual techno-cultural festival.
- Overseeing competition planning, outreach, rulebook finalization, and coordination across multiple organizing teams to ensure smooth execution.

Competitions Coordinator | Elan & Vision, IIT Hyderabad *(2025-2026)*

- Collaborated with a team of **8 coordinators** to organize competitions, independently managing **4 flagship events**, including **BreakFree** and **Glitz & Glamour**.
- Prepared rulebooks, conducted outreach to **250+ colleges**, and coordinated **600+ participants**, **10+ judges**, and on-ground event operations.

Public Relations, Outreach & Marketing Coordinator | Finance & Consulting Club (FCC), IIT Hyderabad *(2025-2026)*

- Worked with a team of **8 coordinators** to promote **Horizon**, the club's flagship event, through outreach to **finance and business schools**.
- Managed institutional communications and marketing efforts, contributing to **150+ registrations/participants.**

RELEVANT COURSES

Introduction to Programming, Data Structures and its Applications, Operating Systems

SCHOLASTIC ACHIEVEMENTS

- Awarded an **A grade in Probability**
- Awarded an **A grade in Statistics**

EXTRACURRICULAR

Secured **3rd place** in the Duo Carrom event at **DIESTA'26,** the annual **inter-departmental sports and cultural** championship at **IIT Hyderabad.**

Secured **3rd place** in the Duo Carrom event at **DIESTA'25**.